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Experiment: 8 Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim

To analyze the effect of steering angle variation in a MIMO beamforming system by studying

steering angle, array gain, and beam direction error using MATLAB.

Objectives

1. To analyze the steering angle of a MIMO beamforming system by directing the beam

towards different target directions using MATLAB.

2. To compare the array gain achieved for different steering angles in a MIMO beamforming

system using MATLAB.

3. To evaluate the beam direction error between the desired steering angle and the actual

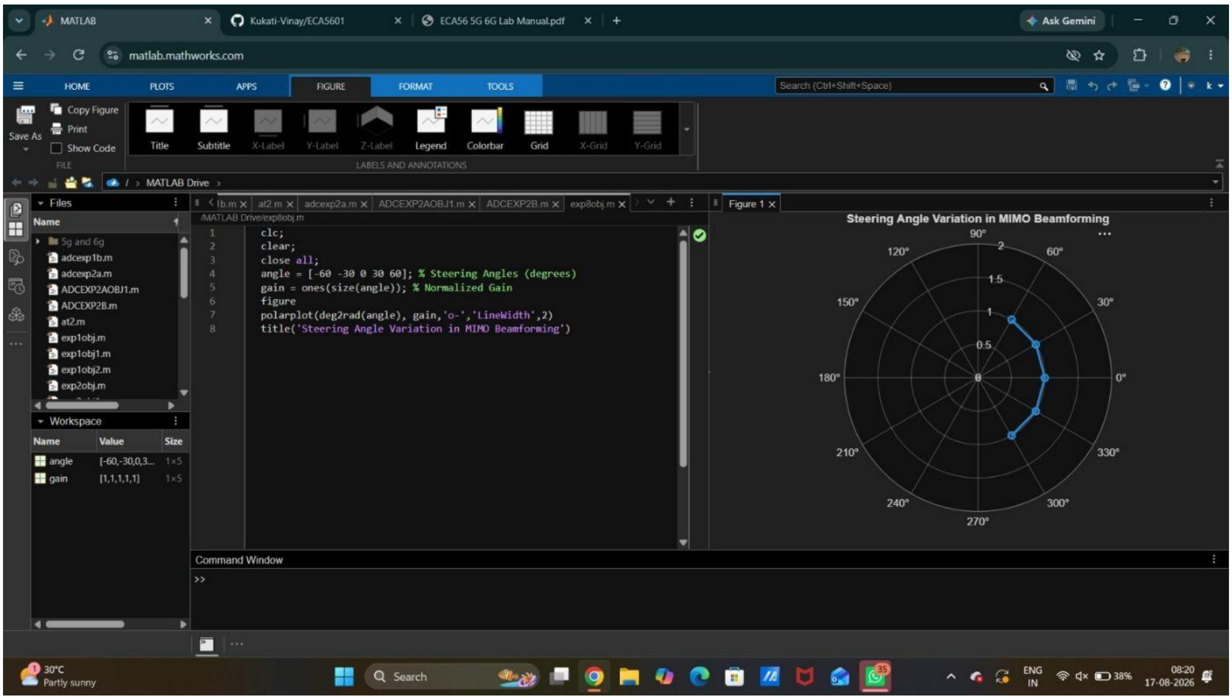
beam direction using MATLAB.

Objective1:

Program:

```
clc; clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = ones(size(angle)); % Normalized Gain

figure

polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')
```

output:

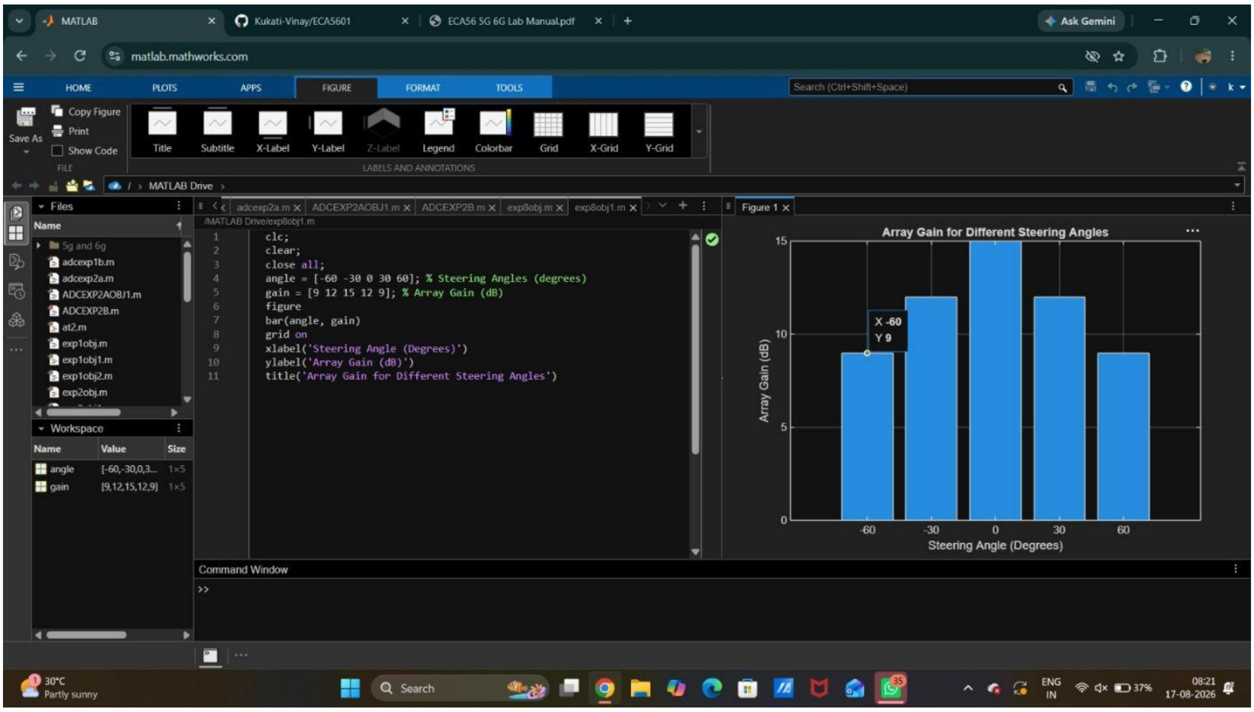


Objective2:

Program:

```
clc; clear;
close all;
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
gain = [9 12 15 12 9]; % Array Gain (dB)
figure
bar(angle, gain)
grid on
xlabel('Steering Angle (Degrees)')
ylabel('Array Gain (dB)')
title('Array Gain for Different Steering Angles')
```

Output:

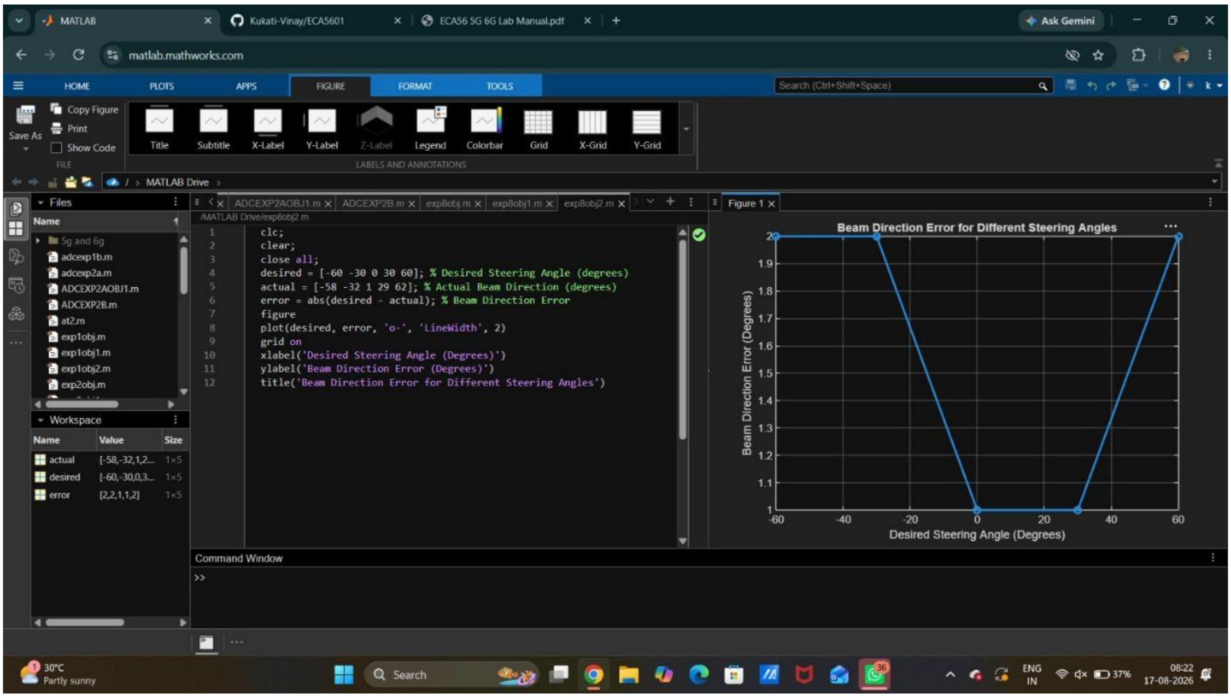


Objective3:

Program:

```
clc;  clear;
close all;
desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees)
actual = [-58 -32 1 29 62]; % Actual Beam Direction (degrees)
error = abs(desired - actual); % Beam Direction Error
figure
plot(desired, error, 'o-', 'LineWidth', 2)
grid on
xlabel('Desired Steering Angle (Degrees)')
ylabel('Beam Direction Error (Degrees)')
title('Beam Direction Error for Different Steering Angles')
```

output:



Result

The MATLAB analysis successfully demonstrated steering angle variation in a MIMO beamforming system. The results showed that beam steering effectively directs the transmitted energy toward different angles, while maintaining high array gain and minimizing beam direction error.